

Problem 1.12

Suppose that the annual interest rate changes from one year to the next. Let i_1 be the interest rate for the first year, i_2 the interest rate for the second year, $\dots$, i_n the interest rate for the n th year. What will be the amount value of an investment of P at the end of the n th year?

Solution

At the end of the first year, the investment grows to

$$P(1 + i_1).$$

At the end of the second year, this amount grows to

$$P(1 + i_1)(1 + i_2).$$

Continuing in this way, at the end of the n th year the amount value is

$$P(1 + i_1)(1 + i_2)(1 + i_3) \cdots (1 + i_n).$$

Equivalently, this can be written in product notation as

$$P \prod_{k=1}^n (1 + i_k).$$